

Homework 5

Arham Lodha

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Problem 1.

Proof. Let $S = \{x \in \mathbb{R} \mid x \geq 2\}$. $\forall x \in S$, by definition of S , $x \geq 2$. Thus by definition of a lower bound, 2 is a lower bound. $\forall y \in \mathbb{R} \ni y$ is a lower bound of S . By definition of a lower bound, $\forall s \in S$, $y \leq s$ thus $y \leq 2$. Thus 2 is the greatest lower bound. Thus by definition of the infimum, $\inf(S) = 2$.

Assume the contrary $\exists t \in \mathbb{R} \ni t$ is an upper bound of S . Then $\forall s \in S$, $t \geq s$, $2 \in S$, thus $t \geq 2$. Thus $t \in S$. $\forall s \in S$, $t + 1 > t \geq s$ and since $t \geq 2$, $t + 1 > 2$. Thus $t + 1 \in S$, however since t is an upper bound of S , $t \geq t + 1$, but $t + 1 > t$. Thus a contradiction occurs, and $\nexists t \in \mathbb{R} \ni t$ is an upper bound. $\square$

Problem 2.

Proof. Let $S = \{x \in \mathbb{R} \mid x > 1\}$. $\forall y \leq 1$, by definition of the set, $\forall s \in S$, $s > 1$ thus $y \leq 1 < s \implies y \leq s$, thus y is by definition a lower bound of S . So any number less than or equal to 1 is a lower bound of S , thus an example lower bound is 0, since $0 \leq 1$.

$1 \leq 1$, thus 1 is a lower bound of S . $\forall n \in \mathbb{R} \ni n$ is a lower bound of S . Assume the contrary, $n > 1$. Then $n \in S$, by definition of S . Let $m = n - \frac{n-1}{2}$. $m - 1 = n - 1 - \frac{n-1}{2} = 2n - n - 2 - 1 = n - 1$, since $n > 1$, thus $m - 1 = n - 1 > 0$. Thus $m > 1$. Thus $m \in S$. $n - m = n - n - \frac{n-1}{2} = \frac{n-1}{2}$, since $n > 1$, then $\frac{n-1}{2} > 0$. Thus $n - m > 0$, thus $n > m$. However by definition of a lower bound, since $m \in S$, $n \leq m$. Thus a contradiction occurs and $n \leq m$ and $n > m$. Thus by contradiction, $n \leq 1$. Since 1 is a lower bound, and all lower bounds are less than it, 1 is a greatest lower bound. $\square$

Problem 3.

Proof. Suppose S , a set, is bounded above. $-S := \{-s : s \in S\}$. Since S is bounded above, $\exists M \in \mathbb{R} \ni \forall s \in S, M \geq s$. By properties of ordering, $-M \leq -s$, by definition of $-S$, $-s \in -S$. Thus $\forall s \in -S$, by definition of $-S$, $-s \in S$ since $-(-s) = s \in -S$, by Algebraic Properties of Reals. By definition of upper bound, $M \geq -s$, by algebraic properties of ordering, $-M \leq s$, thus by definition of a lower bound, $-M$ is a lower bound. By definition of bounded below, $-S$ is bounded below.

Since $-S$ is bounded below, $\exists \inf(-S) = x \in \mathbb{R}$. Since S is bounded above, there exists upper bounds for S , thus $\forall y \in \mathbb{R} \ni y$ is an upperbound of S . Thus $\forall s \in S$, $y \geq s$. By algebraic properties of ordering, $-y \leq -s \in -S$. By definition of $-S$, $\forall t \in -S$, $-y \leq t$. Thus $-y$ is a lower bound of $-S$. By definition of infimum, $-y \leq x$. Thus, $\forall s \in -S$, $-y \leq x \leq s$. $\forall t \in S$, $-t \in -S$, thus $-y \leq x \leq -t$. By algebraic properties of ordering, $y \geq -x \geq t$. Thus $\forall t \in S$, $-x \geq t$, thus $-x$ is an upperbound for S . Moreover, since for all upper bounds of S , y , $y \geq -x$. Thus $-x$ is the least upper bound of S , thus $-x = \sup(S)$. Thus $\inf(-S) = x = -\sup(S)$. Thus $\inf(S) = -\sup(-S)$. $\square$

Problem 4.

Proof. $S := \left\{ 1 - \frac{(-1)^n}{n} : n \in \mathbb{N} \right\}$

Let $x = 2$. $\forall n \in \mathbb{N}$, $2 - 1 + \frac{(-1)^n}{n} = 1 + \frac{(-1)^n}{n}$. $\forall n \in \mathbb{N}$, $\frac{1}{n} \leq 1$. If n is odd, then $1 - \frac{1}{n} > 0$. Thus $2 - [1 - \frac{(-1)^n}{n}] = 2 - 1 + \frac{(-1)^n}{n} = 1 - \frac{1}{n} \geq 0$. Thus $x \geq 1 - \frac{(-1)^n}{n}$. If n is even, thus $1 + \frac{(-1)^n}{n} = 1 + \frac{1}{n} \geq 0$, since sum of 2 positive numbers is positive. Thus $2 - [1 - \frac{(-1)^n}{n}] = 2 - 1 + \frac{(-1)^n}{n} \geq 0$. Thus $x \geq 1 - \frac{(-1)^n}{n}$. Thus $\forall n \in \mathbb{N}$, $x \geq 1 - \frac{(-1)^n}{n}$. Thus by definition of S, $\forall s \in S$, $2 \geq s$. Thus $x = 2$ is an upper bound of S. $\forall n < 2$, $\forall n < 2$. $2 = 1 - \frac{(-1)^1}{1} \in S$. Thus n is not an upper bound of S. Thus $x = 2 = \sup(S)$.

Let $y = \frac{1}{2}$, $\forall n \in \mathbb{N}$, $\frac{1}{2} - [1 - \frac{(-1)^n}{n}] = \frac{1}{2} - 1 + \frac{(-1)^n}{n} = -\frac{1}{2} + \frac{(-1)^n}{n}$. For n odd, $-\frac{1}{2} + \frac{(-1)^n}{n} = -\frac{1}{2} - \frac{1}{n} < 0$, since the sum of 2 negative numbers is negative. For n even, $-\frac{1}{2} + \frac{(-1)^n}{n} = -\frac{1}{2} + \frac{1}{n}$. Since n is even, $n \geq 2$. $\forall n \in \mathbb{N} \ni n \geq 2$, $\frac{1}{2} - \frac{1}{n} \geq 0 \implies \frac{1}{n} - \frac{1}{2} \leq 0$. Thus $-\frac{1}{2} + \frac{(-1)^n}{n} = -\frac{1}{2} + \frac{1}{n} \leq 0$. Thus $\forall n \in \mathbb{N}$, $\frac{1}{2} - [1 - \frac{(-1)^n}{n}] \leq 0 \implies \frac{1}{2} \leq 1 - \frac{(-1)^n}{n}$. Thus by definition of S, $\forall s \in S$, $\frac{1}{2} \leq s$.

Thus $\frac{1}{2}$ is a lower bound of S, $\forall n \in \mathbb{R} \ni n > \frac{1}{2}$. $\frac{1}{2} = 1 - (-1)^2/2$, thus $0.5 \in S$. Since $n > 0.5$, n is not a lower bound of S. Thus since there is no lower bound greater than $\frac{1}{2}$, $\frac{1}{2}$ is the greatest lower bound and thus $\inf(S) = \frac{1}{2}$. $\square$

Problem 5.

Proof. $S := \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$ Let $x = 0$. $\forall n \in \mathbb{N}$, $0 - \frac{1}{n} = -\frac{1}{n} < 0$ since $\frac{1}{n}$ is positive (division of 2 positive values is positive). Thus $0 \leq \frac{1}{n}$. By definition of S, $\forall s \in S$, $0 \leq s$. Thus 0 is a lower bound of S.

$\forall y \in \mathbb{R} \ni y > 0$. Let $n = \lceil \frac{1}{y} \rceil + 1$. Thus $\frac{1}{n} < y$. $n \in \mathbb{N}$, thus $\lceil \frac{1}{y} \rceil + 1 = s \in S$. Thus $y > s$. Thus y is not a lower bound of S. Thus $\forall y \in \mathbb{R} \ni y > 0$, y cannot be a lower bound for S. Thus $x = 0$ is the greatest lower bound of S. Thus $\inf(S) = 0$. $\square$

Problem 6a.

Proof. Given $\epsilon > 0$. Choose $K \in \mathbb{N} \ni K > \frac{1}{\epsilon}$, thus $\epsilon > \frac{1}{K}$. Thus $\forall n \in \mathbb{N} \ni n \geq K$, $\frac{1}{n} \leq \frac{1}{K}$, because $n \geq K$. Thus $\frac{1}{n} < \epsilon$. Since $n^2 + 2 > n^2$, since $n > 0 \implies n^2 > 0$. Thus $\frac{1}{n^2+2} < \frac{1}{n^2}$.

Thus

$$\left| \frac{n}{n^2+2} - 0 \right| = \left| \frac{n}{n^2+2} \right| < \left| \frac{n}{n^2} \right| = \left| \frac{1}{n} \right| \leq \left| \frac{1}{K} \right| < \epsilon.$$

Thus for $\epsilon > 0$, $\exists K \in \mathbb{N} \ni \forall n \in \mathbb{N} \ni n \geq K$, $\left| \frac{n}{n^2+2} - 0 \right| < \epsilon$, thus by definition of the limit of a sequence,

$$\lim \left(\frac{n}{n^2+2} \right) = 0$$

$\square$

Problem 6a.

Proof. Given $\epsilon > 0$. Choose $K \in \mathbb{N} \ni K > \frac{9}{2\epsilon}$, thus $\epsilon > \frac{9}{2K}$. Thus $\forall n \in \mathbb{N} \ni n \geq K$, $\frac{9}{2n} \leq \frac{9}{2K}$, because $n \geq K$. Thus $\frac{9}{2n} < \epsilon$.

Thus

$$\left| \frac{4n+1}{2n+5} - 2 \right| = \left| \frac{4n+1-4n-10}{2n+5} \right| = \left| \frac{-9}{2n+5} \right| = 9 \left| \frac{1}{2n+5} \right|$$

Since $2n+5 > 2n$ because n is positive, then $\frac{1}{2n+5} < \frac{1}{2n}$. And since $2n+5 > 0$, $\frac{1}{2n+5} > 0 \implies \left| \frac{1}{2n+5} \right| = \frac{1}{2n+5}$

Thus, $9 \left| \frac{1}{2n+5} \right| \leq \frac{9}{2n} \leq \frac{1}{2K} < \epsilon$.

Thus for $\epsilon > 0$, $\exists K \in \mathbb{N} \ni \forall n \in \mathbb{N} \ni n \geq K$, $\left| \frac{4n+1}{2n+5} - 2 \right| < \epsilon$, thus by definition of the limit of a sequence,

$$\lim \left(\frac{4n+1}{2n+5} \right) = 2$$

□

Problem 6c.

Proof. Given $\epsilon > 0$. Choose $K \in \mathbb{N} \ni K > \frac{1}{\sqrt{\epsilon}}$, thus $\epsilon > \frac{1}{K^2}$. Thus $\forall n \in \mathbb{N} \ni n \geq K$, $\frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{K}}$, because $n \geq K$. Thus $\frac{1}{\sqrt{n}} < \epsilon$.

Thus

$$\left| \frac{\sqrt{n}}{n+2} - 0 \right| = \left| \frac{\sqrt{n}}{n+2} \right|$$

Since $n+2 > n$, $\frac{\sqrt{n}}{n+2} < \frac{\sqrt{n}}{n}$. And since, $\frac{\sqrt{n}}{n+2} > 0$, since $n > 0$ so $n+2 > 0$. Thus,

$$\left| \frac{\sqrt{n}}{n+2} - 0 \right| = \left| \frac{\sqrt{n}}{n+2} \right| < \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} < \frac{1}{\sqrt{K}} < \epsilon$$

Thus for $\epsilon > 0$, $\exists K \in \mathbb{N} \ni \forall n \in \mathbb{N} \ni n \geq K$, $\left| \frac{\sqrt{n}}{n+2} - 0 \right| < \epsilon$, thus by definition of the limit of a sequence,

$$\lim \left(\frac{\sqrt{n}}{n+2} \right) = 0$$

□